

Matching Division Stories, Drawings, and Equations

Answer Key

Grades 2–3 Operations and Algebraic Thinking

Quick Answers

1. 4	4. 4	7. 4	10. 9
2. 5	5. 3	8. 4	11. 5
3. 6	6. 6	9. 7	12. 10

Curriculum

Level: Grades 2–3 Operations and Algebraic Thinking

3.OA.A.3 / 4.OA.A.2 – *problems 1, 2, 4, 5, 6, 8, 9, 10, 12*

3.OA.C.7 – *problems 3, 7, 11*

Difficulty 1–5 of 5 across 12 tagged checks.

Common wrong answers

4. *If they got 18: subtracted the two numbers instead of dividing*

8. *If they got 256: multiplied the two numbers instead of dividing*

10. *If they got 4: counted the rows as the answer instead of dividing to find the number in each row*

12. *If they got 2: divided one pack by the tables and forgot the other packs*

1. 12 stickers shared by 3 friends.

Step 1. Name the three parts before writing anything. The total in all is 12 and the number of groups is 3; what is unknown is how many are in each group. That is the sharing shape of division.

Step 2. Write it as total $\div$ groups: $12 \div 3$, and count out 12 into 3 equal piles — 4 in each pile. 4

Check with multiplication: $3 \times 4 = 12$, which is the same fact read the other way.

2. The picture of 20 counters in 4 equal groups.

Step 1. A drawing already shows two of the three parts: count the total (20 counters) and count the boxes (4 groups). The unknown is the size of one group.

Step 2. $20 \div 4$: each box holds the same number, and counting one box gives 5. 5

Matching multiplication: $4 \times 5 = 20$ — 4 groups of 5.

3. $3 \times \square = 18$ and its division partner.

Step 1. A missing *factor* is exactly what division finds, so turn the multiplication around instead of guessing.

Step 2. The partner sentence is $18 \div 3 = \square$. Skip-count by threes to 18: 3, 6, 9, 12, 15, 18 — that is 6 threes. 6

Fact family: $3 \times 6 = 18$, $6 \times 3 = 18$, $18 \div 3 = 6$, $18 \div 6 = 3$.

4. 24 crayons packed 6 to a box.

Step 1. This story gives the total (24) and the size of each group (6 in a box). The unknown is the *number* of groups, so it is the measurement shape of division, not the sharing shape.

Step 2. $24 \div 6$: take 6 away at a time — 24, 18, 12, 6, 0. That is 4 boxes. 4

Watch out: a student who reads “24 and 6” without reading the story may subtract and write 18. Subtraction takes 6 away *once*; division takes 6 away until nothing is left.

5. Which story matches $15 \div 5 = \square$?

Step 1. Read the equation out loud first: “15 in all, split into 5 equal parts.” Now look for the story that starts with 15 as the total.

Step 2. Story **A** is the match — 15 shells split into 5 equal piles. Story B makes 15 the size of each group, so its total would be much larger than 15. Answering A: $15 \div 5 = 3$ shells in each pile. 3

Language clue: “put into 5 equal piles” names groups; “15 in each” names group size.

6. The picture of 18 counters in 3 equal groups.

Step 1. Read the same two facts off the drawing: 18 counters in all, 3 boxes. The unknown is the size of one group.

Step 2. $18 \div 3$: each box holds the same amount, and one box holds 6. 6

Matching multiplication: $3 \times 6 = 18$. The same picture tells both sentences — multiplication puts the groups together, division breaks them apart.

7. The fact family of 7, 4 and 28.

Step 1. A missing factor again, so use the division partner rather than guessing and checking.

Step 2. $28 \div 7$: count by sevens — 7, 14, 21, 28. That is 4 sevens. 4

All four sentences: $7 \times 4 = 28$, $4 \times 7 = 28$, $28 \div 7 = 4$, $28 \div 4 = 7$. Every one uses the same three numbers.

8. 32 apples in baskets of 8.

Step 1. Decide which number is the total before writing the equation. “A farm stand has 32 apples” is the whole amount; “8 apples in each basket” is the group size. The total is always the number you divide.

Step 2. $32 \div 8$: count by eights — 8, 16, 24, 32. That is 4 baskets. 4

Watch out: multiplying gives 256, which is more apples than the stand ever had. If an answer is bigger than the total, the operation was wrong.

9. Which story matches $42 \div 6 = \square$?

Step 1. Say the equation in words: “42 in all, put into groups of 6.” The total must be 42.

Step 2. Story **A** matches — 42 muffins with 6 on each tray. Story B says 42 muffins are on *each* of 6 trays, so 42 is a group size there and the total would be far more than 42. Answering A: $42 \div 6 = 7$ trays. 7

Why B fails: B describes multiplication, 6×42 , not division.

10. The array of 36 counters in 4 rows.

Step 1. An array gives the total (36) and the number of rows (4). The question asks for the size of a row, so divide the total by the number of rows.

Step 2. $36 \div 4$: count by fours to 36 — 4, 8, 12, 16, 20, 24, 28, 32, 36. That is 9, so each row holds 9 counters. 9

The trap: answering 4 would be reporting the number of *rows*, which the picture already told you. Read the question, then decide which part of the array is unknown.

11. $45 \div \square = 9$ — the divisor is missing.

Step 1. Here the total (45) and the answer (9) are known, so rewrite the sentence as multiplication: $\square \times 9 = 45$.

Step 2. Count by nines — 9, 18, 27, 36, 45 — which takes 5 nines. 5

A story it could tell: 45 marbles are shared equally among 5 children, and each child gets 9.

12. Challenge: 5 packs of 6 pencils shared by 3 tables.

Step 1. You cannot divide yet, because the total is not given — only the packs are. Find the total first, then share it.

Step 2. Total pencils: $5 \times 6 = 30$.

Step 3. Now share the total among the tables: $30 \div 3 = 10$ pencils for each table. 10

The trap: dividing $6 \div 3$ gives 2 and only shares *one* pack. A picture of 3 groups with 10 pencils in each shows the whole 30.